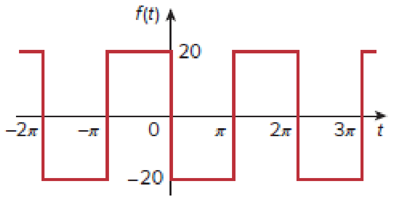
Chapter 17, Problem 3PP

Problem

Find the Fourier series of the function *f*(*t*) in Fig. 17.14.

Figure 17.14:



Step-by-step solution

Step 1 of 5

Refer to wave in Figure 17.14 in the textbook.

The function $f(t)$ is,

$$f(t)=\begin{cases}-20, & 0<t<\pi \\ 20, & \pi<t<2\pi\end{cases}$$

Time period of $f(t)$ is,

$$T=2\pi$$

Frequency of $f(t)$ is,

$$\begin{aligned}\omega_0&=\frac{2\pi}{T}\\&=\frac{2\pi}{2\pi}\\&=1\end{aligned}$$

Step 2 of 5

Calculation of a_0.

$$\begin{aligned}a_0&=\frac{1}{T}\int_0^T f(t)\,dt\\&=\frac{1}{2\pi}\left[\int_0^{\pi} f(t)\,dt+\int_{\pi}^{2\pi} f(t)\,dt\right]\\&=\frac{1}{2\pi}\left[\int_0^{\pi}(-20)\,dt+\int_{\pi}^{2\pi}20\,dt\right]\\&=\frac{20}{2\pi}\left[-t\Big|_0^{\pi}+t\Big|_{\pi}^{2\pi}\right]\\&=\frac{10}{\pi}\left[(-\pi+0)+(2\pi-\pi)\right]\\&=\frac{10}{\pi}\left[-\pi+\pi\right]\\&=0\end{aligned}$$

Therefore, the value of a_0 is 0.

Step 3 of 5

Calculation of a_n.

$$\begin{aligned}
a_n &= \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt \\
&= \frac{2}{2\pi}\int_0^{2\pi} f(t)\cos nt\,dt \\
&= \frac{1}{\pi}\left[\int_0^{\pi} f(t)\cos nt\,dt + \int_{\pi}^{2\pi} f(t)\cos nt\,dt\right] \\
&= \frac{1}{\pi}\left[\int_0^{\pi} (-20)\cos nt\,dt + \int_{\pi}^{2\pi} 20\cos nt\,dt\right] \\
&= \frac{20}{\pi}\left[-\int_0^{\pi} \cos nt\,dt + \int_{\pi}^{2\pi} \cos nt\,dt\right] \\
&= \frac{20}{\pi}\left[-\frac{\sin nt}{n}\bigg|_0^{\pi} + \frac{\sin nt}{n}\bigg|_{\pi}^{2\pi}\right] \\
&= \frac{20}{\pi}\left[\frac{-1}{n}(\sin n\pi - \sin 0n\pi) + \frac{1}{n}(\sin 2n\pi - \sin n\pi)\right] \\
&= \frac{20}{\pi}\left[\frac{-1}{n}(0) + \frac{1}{n}(0)\right] \\
&= 0
\end{aligned}$$

Therefore, the value of a_n is 0.

Step 4 of 5

Calculation of b_n.

$$\begin{aligned}
b_n &= \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt \\
&= \frac{2}{2\pi}\int_0^{2\pi} f(t)\sin nt\,dt \\
&= \frac{1}{\pi}\left[\int_0^{\pi} f(t)\sin nt\,dt + \int_{\pi}^{2\pi} f(t)\sin nt\,dt\right] \\
&= \frac{1}{\pi}\left[\int_0^{\pi} (-20)\sin nt\,dt + \int_{\pi}^{2\pi} 20\sin nt\,dt\right] \\
&= \frac{20}{\pi}\left[-8\int_0^{\pi} \sin nt\,dt + 8\int_{\pi}^{2\pi} \sin nt\,dt\right] \\
&= \frac{20}{\pi}\left[\frac{\cos nt}{n}\bigg|_0^{\pi} - \frac{\cos nt}{n}\bigg|_{\pi}^{2\pi}\right] \\
&= \frac{20}{2\pi}\left[(\cos n\pi - \cos 0n\pi) - (\cos 2n\pi - \cos n\pi)\right] \\
&= \frac{20}{n\pi}\left[(-1)^n - 1\right] - \frac{20}{n\pi}\left[1 - (-1)^n\right] \\
&= \frac{20}{n\pi}\left((-1)^n - 1 - 1 + (-1)^n\right)
\end{aligned}$$

Step 5 of 5

Continue the calculations.

$$= \frac{40}{n\pi}\left((-1)^n - 1\right)$$

$$= \begin{cases} -\dfrac{80}{n\pi} & \text{for } n \text{ odd,} \\ 0 & \text{for } n \text{ even.} \end{cases}$$

Therefore, the value of b_n is $-\frac{80}{n\pi}$ for n odd and 0 for n even.

The function $f(t)$ is,

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos nw_0 t + b_n \sin nw_0 t\right)$$

Substitute 0 for a_0, 0 for a_n and $-\frac{80}{n\pi}$ and 0 for b_n in equation (1).

$$f(t) = 0 + \sum_{k=1}^{n}\left[0\cos nt + \left(-\frac{80}{n\pi} + 0\right)\sin nt\right]$$

$$= -\frac{80}{\pi}\sum_{k=1}^{n}\frac{1}{n}\sin nt \quad ,n \text{ is odd}$$

Therefore, the Fourier series of the function $f(t)$ is $\boxed{-\frac{80}{\pi}\sum_{k=1}^{\infty}\frac{1}{n}\sin nt, \quad n = 2k-1}$.